Hang Du

Ph.D. Candidate in Software Engineering, University of California, Irvine

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Education

University of California, Irvine 2022 – Present (Expected 2028)

Ph.D. in Software Engineering

Advisor: James A. Jones

University of California, Irvine 2021 – 2025

Master of Science in Software Engineering

Advisor: James A. Jones (Transferred to Ph.D. program Sep 2022; conferred Dec 2025)

Coursework: Program Analysis, Software Architecture, Software Testing, Information Retrieval

McGill University, Montreal, Canada Feb 2020

Machine Learning Certificate Program

Coursework: Computational Applied Statistics, Practical Machine Learning

Jilin University, China 2017 – 2021

Bachelor of Science in Computer Science and Technology

Bachelor of Finance (minor)

Peer-Reviewed Publications

[1] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2026. *How Does Killing Surviving Mutants Help Detect Real Bugs with Assertion Generation: A Controlled Experiment*. In Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026) [PDF].

ACM SIGSOFT Distinguished Paper Award (19 of 888 submissions)

Direct Acceptance Rate: 10.1%

[2] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2026. *Revealing Regressions: A Comparative Study of State-Capture Strategies in Validating Program Behavior*. Proceedings of the ACM on Software Engineering (FSE 2026); also in the journal Proceedings of the ACM on Software Engineering (PACMSE) [PDF].

Direct Acceptance Rate: 9.5%

[3] Monil Narang, **Hang Du**, and James A. Jones. 2025. *What's DAT Smell? Untangling and Weaving the Disjoint Assertion Tangle Test Smell*. In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025) [PDF].

Acceptance Rate: 20.6%

[4] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2025. *Leveraging Propagated Infection to Crossfire Mutants*. In Proceedings of the 47th IEEE/ACM International Conference on Software Engineering (ICSE 2025) [PDF].

Oral presentation in Ottawa, Canada (presented by Hang Du). Acceptance Rate: 21.4%

[5] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2024. *Ripples of Mutation: An Empirical Study of Propagation Effects in Mutation Testing*. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (ICSE 2024) [PDF].

Oral presentation in Lisbon, Portugal (presented by Hang Du). Acceptance Rate: 21.2%

[6] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2023. *To Kill a Mutant: An Empirical Study of Mutation Testing Kills*. In Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2023) [PDF].

Oral presentation in Seattle, USA (presented by Hang Du). Acceptance Rate: 25.9%

Selected Projects

Mutation testing is a technique that evaluates a test suite by repeatedly injecting one artificial fake bug into a program at a time and checking whether the tests catch it; here, a "mutation" is one such injected bug and a "mutant" is the resulting buggy copy of the program.

A Decade of Mutation Testing in the Wild: Practice, Mindset, and Community (ongoing)

- Studying how open-source projects adopt and sustain mutation testing over ten years, characterizing the **testing mindsets and community practices** that form around artificially injected bugs.
- Analyzing the full mutation-testing history of long-running projects such as **Checkstyle**, which has practiced mutation testing for **a decade** and accumulated **over 36,000 artificially injected bugs**.
- Tracing how the practice is evolving: recent projects such as n8n now **delegate mutation testing to AI agents** and enforce **mutation score as a quality-assurance gate** for both human- and AI-written code.
- Deriving insights into how developers reason about **test adequacy**, which mutants they choose to address, and how these decisions evolve as projects mature.

Test Suite Improvement with Assertion Amplification and Mutant Killing

- Proposed a **test-failure taxonomy** of the runtime symptoms through which tests detect mutants and real bugs, and built a **Java analysis framework** that traces how a fault **progressively infects program state and propagates** to a failure; evaluated at scale on artificially injected faults and real bugs from Defects4J (ISSTA 2023, ICSE 2024).
- Turned those propagation insights into an automated approach that generates **assertion candidates** for human-written test suites and optimizes them to **"crossfire" multiple hidden faults** at once, detecting all detectable faults with only **1.1% of candidates** and boosting fault detection by **6**× on average across ten open-source Java projects and 1.2 million test executions; supporting theoretical model and artifact released (ICSE 2025).
- Ran a **controlled experiment** on whether **killing surviving mutants** actually helps detect **unseen real bugs**: of 642 Defects4J bugs, **104** could have been caught by **one localized assertion** added to an existing passing test (13,266 validated augmentation options generated), and 63 of those were reachable by killing Major mutants with the technique.
- Showed that the benefit does not require exhaustive killing—randomly killing **30%** of relevant surviving mutants, one assertion each, detects **84.5%** of those bugs—and explained the **mechanism**: mutation-derived assertions validate program outputs that **overlap with, and are often subsets of**, the outputs corrupted by real bugs, while real bugs corrupt larger program state than mutants (ISSTA 2026, **ACM SIGSOFT Distinguished Paper Award**).

Technical Skills

Program instrumentation & dynamic analysis: Java bytecode instrumentation, runtime state capture and profiling, fault-propagation tracing, mutation analysis pipelines at scale (1.2M+ test executions), Defects4J

AST-based code transformation: static analysis and source-to-source refactoring with JavaParser, automated test-smell detection and repair, assertion synthesis and amplification

Software testing: JUnit, mocking/stubbing and testable design, test adequacy and mutation scoring, large-scale experiment harnesses, controlled experiments and developer studies

AI for software engineering: LLM-augmented developer tools, AI agents for testing and code review, evaluation of generated tests and code, Model Context Protocol (MCP)

Languages & tooling: Java (primary), Python; Git, Docker, Linux, continuous integration

Teaching & Mentoring

DTEI Summer Graduate Scholar, University of California, Irvine Jul 2026 – Sep 2026

- Awarded a competitive $5,000 fellowship through the Division of Teaching Excellence and Innovation (DTEI) at UC Irvine.
- Collaborated with faculty mentors to transform INF 43: Introduction to Software Engineering into a project-based course, integrating AI agents and generative AI tools into software engineering education and course projects.
- Completed a 200-hour teaching-development program focused on inclusive pedagogy, accessibility, instructional innovation, and generative AI in education.

Pre-Professor Program (PREPP) Fellow, California State University Aug 2026 – Dec 2026

- Awarded a competitive $1,000 fellowship through the CSU Pre-Professor Program (PREPP).
- Selected to participate in a faculty-preparation program designed to introduce doctoral students to academic careers within the California State University system.
- Engaged in faculty mentorship, professional development, and academic career exploration focused on teaching, research, service, and student success.

Associate Instructor / Lecturer, University of California, Irvine Jun 2023 – Aug 2023

ICS 45J: Programming in Java

- Served as the sole course instructor, responsible for designing the syllabus, assignments, exams, and grading rubrics for an undergraduate course on object-oriented programming in Java.
- Delivered all lectures and lab sessions, providing instruction on core programming concepts, data structures, and software design principles.
- Supervised and coordinated with Teaching Assistants to manage grading, lab support, and student feedback.
- Achieved strong teaching evaluations (Instructor: 8.08/9.00, Course: 8.12/9.00) reflecting effective instruction and student engagement.

Guest Lecturer, University of California, Irvine Feb 2026

SWE 261P: Software Testing (graduate)

- Enhanced course materials by integrating state-of-the-art industry practices and recent research advances in software testing, particularly in mocking, stubbing, and testable design.
- Developed the slides and assignments for mocking/stubbing and testable design.

Guest Lecturer, University of California, Irvine Apr 2025

INF 215: Dynamic Analysis, Testing, and Debugging (graduate)

- Delivered a special lecture on mutation testing and its applications in assessing test adequacy, revealing fault propagation, and guiding test improvement.

Teaching Assistant, University of California, Irvine Jun 2022 – Present

- INF 122: Software Design: Structure and Implementation — Winter 2026
- INF 43: Introduction to Software Engineering — Fall 2021 (Reader), Winter 2022, Fall 2022, Fall 2023, Fall 2025, Spring 2026
- INF 115: Software Testing, Analysis, and Quality Assurance — Spring 2023, Spring 2024, Fall 2024, Spring

2025

- SWE 261P: Software Testing and Debugging — Winter 2023, Winter 2024, Winter 2025

Graduate Research Mentor, University of California, Irvine Oct 2023 – Jun 2024

- Mentored an undergraduate UROP participant in the development of an honors thesis titled "Generative AI for Mutation Testing."
- Advised on research topic selection, proposal development for UROP funding, and the creation of a research poster for the 2024 Undergraduate Research Symposium.
- Conducted weekly research meetings to provide guidance and feedback.

Graduate Research Mentor, University of California, Irvine Sep 2023 – Jun 2025

- Mentored a master's student on the thesis "Automated Detection and Refactoring of Disjoint-Assertion Tangle: A Novel Test Smell."
- Guided the student through experimental design, implementation, paper writing, and thesis development, resulting in a co-authored paper accepted at ASE 2025.

Industry Experience

Financial Analytics Intern, First Capital Securities Nov 2020 – Dec 2020

Supervisor: Ming Shao

- Built data-driven reports and dashboards to support investment decision-making and client advisory services.

Certificates & Training

Essential Class Design Certificate Program, University of California, Irvine — Winter 2026
Mentoring Excellence Certificate Program, University of California, Irvine — Fall 2025
Public Speaking: Activate to Captivate Certificate Program, University of California, Irvine — Winter 2024
Emotional Intelligence Certificate Program, University of California, Irvine — Spring 2024

Professional Affiliations

- Member, IEEE
- Member, Association for Computing Machinery (ACM)
- Member, ACM Special Interest Group on Software Engineering (ACM SIGSOFT)

Awards & Funding

NSF Travel Award for ISSTA — 2026
DTEI Summer Graduate Scholar, University of California, Irvine — 2026
CSU Pre-Professor Program Fellowship — 2026
Undergraduate Research Experience Fellowship (mentor), University of California, Irvine — 2024
Summer Fellowship Award, Department of Informatics, University of California, Irvine — 2024
ICSE Student Mentoring Workshop Award, funded by NSF — 2024
ACM SIGSOFT CAPS Travel Grant — 2023
Chair's Award, University of California, Irvine — 2022
Academic Scholarship, Jilin University, China — 2018, 2019, 2020

Service

Shadow Program Committee Member, 49th IEEE/ACM International Conference on Software Engineering (ICSE 2027) 2027

Reviewer, ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), Artifact Evaluation Track 2026

Reviewer, IEEE/ACM International Conference on Automated Software Engineering (ASE), Artifact Evaluation Track 2026

Reviewer, IEEE Transactions on Software Engineering (TSE) 2026

Student Volunteer, ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA) 2026

Junior Program Committee Member, International Conference on Mining Software Repositories (MSR) 2026, 2027

- Reviewed and evaluated research papers submitted to the MSR Research/Technical Track.
- Provided constructive feedback to authors and participated in the peer-review and discussion process.

External Reviewer, IEEE Transactions on Reliability 2024

- Reviewed and provided detailed feedback on submitted manuscripts in software reliability and testing.

Graduate Research Mentor, University of California, Irvine Oct 2023 – Jun 2024

- Mentored an undergraduate UROP (Undergraduate Research Opportunities Program) participant in the development of an honors thesis titled “Generative AI for Mutation Testing.”
- Advised on research topic selection, proposal development for UROP funding, and the creation of a research poster for the 2024 Undergraduate Research Symposium.
- Conducted weekly research meetings to provide guidance and feedback.